

Study and Research Focus

Software systems, data-driven systems and embedded real-time software. Applying for a master's programme in Computer Science and related fields, with interest in laboratory research and assistant roles.

Education

2023–2027 **Sharif University of Technology, Tehran, Iran**

B.Sc., Electrical Engineering (Digital Systems focus) • expected Aug 2027

- **Cumulative GPA: 16.88/20.** Above the department average (16.49) and university average (16.09) for this cohort, as printed on the [Sharif transcript](#).
- **Class rank: 90 of 200 students** in the 2023 Electrical Engineering entry cohort (Sharif registrar certificate).
- Most recent semester (Spring 2026): **19.30/20** over 19 units.
- **National rank 134** among about 150,000 candidates in the Mathematics and Physics group of the Iranian University Entrance Exam (Konkur).
- **Relevant coursework (grade/20):** Fundamentals of Programming (20.0); Foundations of Data Science (19.3); Embedded Systems (19.8); Object-Oriented Programming (17.8); Computer Interfacing, a graduate course (19.0); Computer Structure and Microprocessor with lab (17.0). In progress (Fall 2026): Parallel Programming; Data Structures and Algorithms.

Projects

- 2026–present **SUT Music: Data Platform and Recommendation Engine** [source](#)
Foundations of Data Science course, Dr. Salavati • Python, scikit-learn, PostgreSQL, FastAPI, React
- **Data:** Team of 3 (project lead); cleaned a Telegram music community's history into a database of 14,800+ tracks.
 - **Recommenders:** Two generations (matrix factorization; two-stage ensemble) behind a backend API and a Telegram Web App.
- 2024–present **EE Department Bot Ecosystem and BotOS Framework**
Lead developer • Python (asyncio), PostgreSQL, Docker, Google Sheets API
- **System:** Bots for course-file distribution, assignment grading and re-judge ticketing, used by the EE department and two student groups; adopted by a second university (IUST).
 - **Framework:** BotOS deploys new bots from declarative JSON configuration in isolated Docker containers.
- Spring 2026 **Smart Surveillance System on an Embedded Linux Board** [source](#)
Embedded Systems course, Dr. Gholampour • Orange Pi Zero Plus 2 (H5), C, Python
- **System:** Individual project built around an on-board detector, an HTTPS dashboard, a REST API and MQTT/e-mail alerts.
 - **Reliability and evaluation:** Added a watchdog and thermal throttling; measured boot time, detection accuracy and load.
- Spring 2026 **OTP Memory Controller (RTL design, synthesis and verification)** [source](#)
ASIC/FPGA System Design course, Dr. Baharvand • Verilog, Cadence Genus and Xcelium
- **Design:** Co-designed a power-on FSM controller that reads a 256-bit OTP array, applies hardware patches and checks a CRC-16.
 - **Team and flow:** Team of 4, led week 1 (rotating lead); verified with a testbench, then ran synthesis and gate-level simulation.
- Spring 2025 **MIPS32 Processor Implementation on FPGA**
Computer Architecture lab, Dr. Movahedin • Verilog, MIPS32 assembly, Xilinx Zynq-7010
- **Design and validation:** Single-cycle, multi-cycle and pipelined MIPS32 CPUs; the multi-cycle one ran on an FPGA board.

Teaching Experience

- 2024–2026 **Teaching Assistant, Fundamentals of Programming and OOP**
Sharif University of Technology • Dr. Vosoughi Vahdat and Dr. Arasteh • four consecutive semesters
- Fall 2025 **Teaching Assistant, Electronics 1**
Sharif University of Technology • Dr. Fakharzadeh, Dr. Sarvari, Dr. Medi, Dr. Besharatirad
- Fall 2025 **Lab Assistant, Logic Circuit Lab (FPGA)**
Sharif University of Technology • Dr. Movahedin • supported students designing on Zynq-7010 FPGA boards
- Spring 2026 **Teaching Assistant, Theory of Circuits**
Iran University of Science and Technology (IUST)

Industrial Training

- Summer 2026 **Trainee, Parsian Zarin (ABLY): MQL5 expert advisors; Excellent Pass**

Technical Skills

- **Languages:** Python, C, C++, Java, SQL, Verilog, MIPS32 assembly.
- **Software engineering:** backend and REST API development, asynchronous programming, Docker, Git, CI/CD, Linux, relational databases (PostgreSQL, SQLite), React (web front end).
- **Data and machine learning:** data analysis, machine learning, recommender systems.
- **Embedded and hardware:** embedded Linux on single-board computers, FPGA design with Xilinx Vivado, Cadence Genus and Xcelium.

Languages

Persian (native); English (official test score pending).